

(Universidade de São Paulo)
Trabalho 3:
Equações Parabólicas e Hiperbólicas

Métodos Numéricos em Equações Diferenciais

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1 EDP

Da EDP

$$u_t(x, t) + u_x(x, t) = \frac{1}{\mathbb{P}e} u_{xx}(x, t), \quad (x, t) \in [0, 15] \times [0, 12] \quad (1)$$

definimos $x_j = j\Delta x$ e $t_n = n\Delta t$, com $j = 0, 1, \dots, N_x$ e $n = 0, 1, \dots, N_t$, então para $U_j^n \approx u(x_j, t_n)$ discretizamos

2 Discretizações

2.1 Progressivas no tempo e centradas no espaço

$$\begin{aligned} & \frac{u(x_j, t_{n+1}) - u(x_j, t_n)}{\Delta t} + \mathcal{O}(\Delta t) + \frac{u(x_{j+1}, t_n) - u(x_{j-1}, t_n)}{2\Delta x} + \mathcal{O}(\Delta x^2) \\ &= \frac{1}{\mathbb{P}e} \frac{u(x_{j+1}, t_n) - 2u(x_j, t_n) + u(x_{j-1}, t_n))}{\Delta x^2} + \mathcal{O}(\Delta x^2) \end{aligned} \quad (2)$$

que produz o método

$$\frac{U_j^{n+1} - U_j^n}{\Delta t} + \frac{U_{j+1}^n - U_{j-1}^n}{2\Delta x} = \frac{1}{\mathbb{P}e} \frac{U_{j-1}^n - 2U_j^n + U_{j+1}^n}{\Delta x^2} \quad (3)$$

ou, equivalentemente

$$\begin{aligned} U_j^{n+1} &= \left(\frac{\Delta t}{\mathbb{P}e\Delta x^2} + \frac{\Delta t}{2\Delta x} \right) U_{j-1}^n + \left(1 - \frac{2\Delta t}{\mathbb{P}e\Delta x^2} \right) U_j^n + \left(\frac{\Delta t}{\mathbb{P}e\Delta x^2} - \frac{\Delta t}{2\Delta x} \right) U_{j+1}^n \\ &= (\alpha + \beta)U_{j-1}^n + (1 - 2\alpha)U_j^n + (\alpha - \beta)U_{j+1}^n \end{aligned} \quad (4)$$

onde $\alpha = \frac{\Delta t}{\mathbb{P}e\Delta x^2}$ e $\beta = \frac{\Delta t}{2\Delta x}$.

Para as condições de contorno periódicas, definimos $U_{N_x+1}^n = U_0^n$, então:

$j = 0$:

$$U_0^{n+1} = (\alpha + \beta)U_{-1}^n + (1 - 2\alpha)U_0^n + (\alpha - \beta)U_1^n = (1 - 2\alpha)U_0^n + (\alpha - \beta)U_1^n + (\alpha + \beta)U_{N_x}^n$$

$j = N_x$:

$$U_{N_x}^{n+1} = (\alpha + \beta)U_{N_x-1}^n + (1 - 2\alpha)U_{N_x}^n + (\alpha - \beta)U_{N_x+1}^n = (\alpha - \beta)U_0^n + (\alpha + \beta)U_{N_x-1}^n + (1 - 2\alpha)U_{N_x}^n$$

Portanto, para a forma matricial, teremos

$$\mathbf{U}^{n+1} = A\mathbf{U}^n \quad (5)$$

$$\text{onde } \mathbf{U}^n = \begin{bmatrix} U_0^n \\ U_1^n \\ U_2^n \\ \vdots \\ U_{N_x}^n \end{bmatrix} \text{ e } A = \begin{bmatrix} (1 - 2\alpha) & (\alpha - \beta) & 0 & \cdots & (\alpha + \beta) \\ (\alpha + \beta) & (1 - 2\alpha) & (\alpha - \beta) & \cdots & 0 \\ 0 & (\alpha + \beta) & (1 - 2\alpha) & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ (\alpha - \beta) & 0 & \cdots & (\alpha + \beta) & (1 - 2\alpha) \end{bmatrix}$$

Da condição inicial $u(x, 0) = \exp(-20(x - 2)^2) + \exp(-(x - 5)^2)$, obtemos na forma vetorial

$$\mathbf{U}_j^0 = \begin{bmatrix} \exp\{-20(x_0 - 2)^2\} + \exp\{-(x_0 - 5)^2\} \\ \exp\{-20(x_1 - 2)^2\} + \exp\{-(x_1 - 5)^2\} \\ \vdots \\ \exp\{-20(x_{N_x} - 2)^2\} + \exp\{-(x_{N_x} - 5)^2\} \end{bmatrix}$$

2.1.1 Gráficos

Usando 430 pontos igualmente espaçados em x , obtemos $\Delta x = \frac{15}{430 - 1} \approx 3.4965 \cdot 10^{-2}$. Para satisfazer a desigualdade de estritamente menor, utilizamos $\varepsilon = 0.1$ de forma que $\Delta t = (1 - \varepsilon) \min \left\{ \frac{\Delta x^2}{\frac{2}{\mathbb{P}e} + \Delta x}, \Delta x \right\}$, logo:

- $\mathbb{P}e = 0.1 \Rightarrow \Delta t \approx 5.4919 \cdot 10^{-5}$
- $\mathbb{P}e = 1 \Rightarrow \Delta t \approx 5.4070 \cdot 10^{-4}$
- $\mathbb{P}e = 20 \Rightarrow \Delta t \approx 8.1525 \cdot 10^{-3}$

Assim, obtivemos os seguintes resultados:

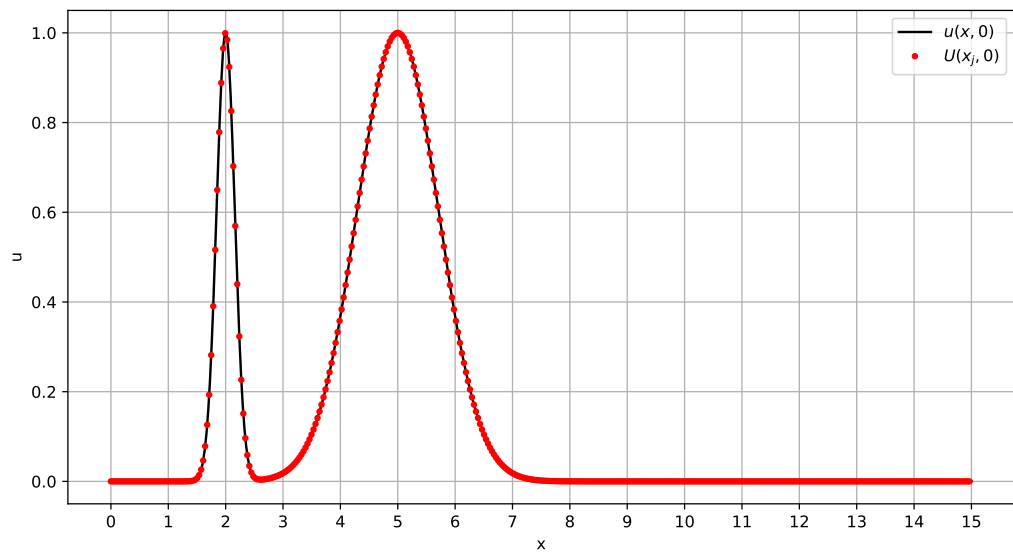


Figura 1: Condição inicial

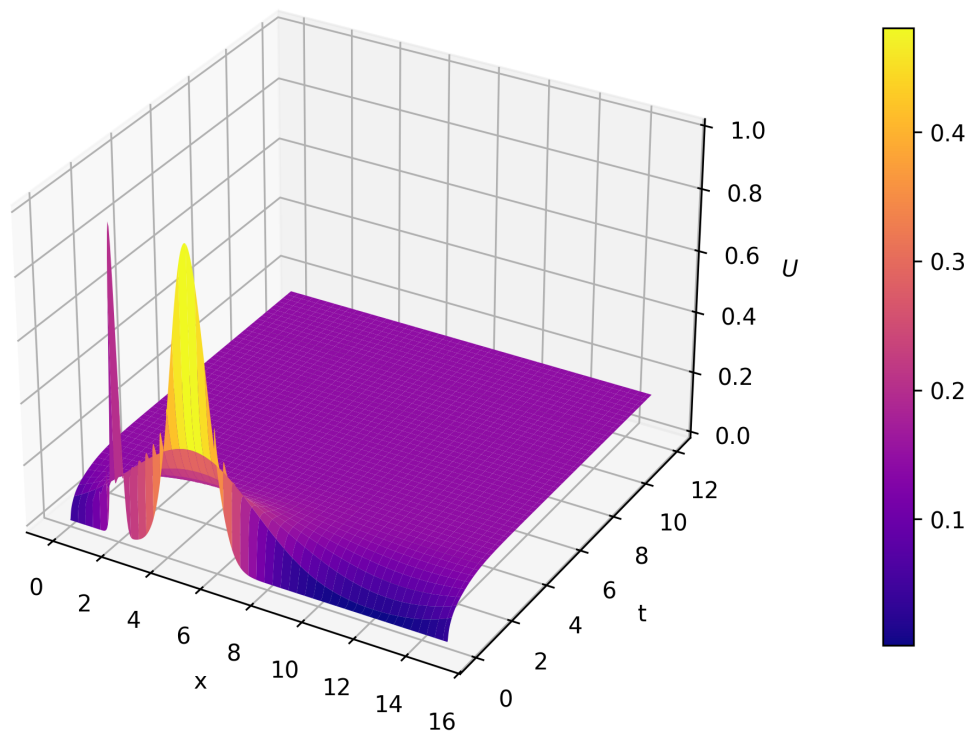


Figura 2: Solução numérica para $Pe = 0.1 \ll 1$ (Difusão dominante)

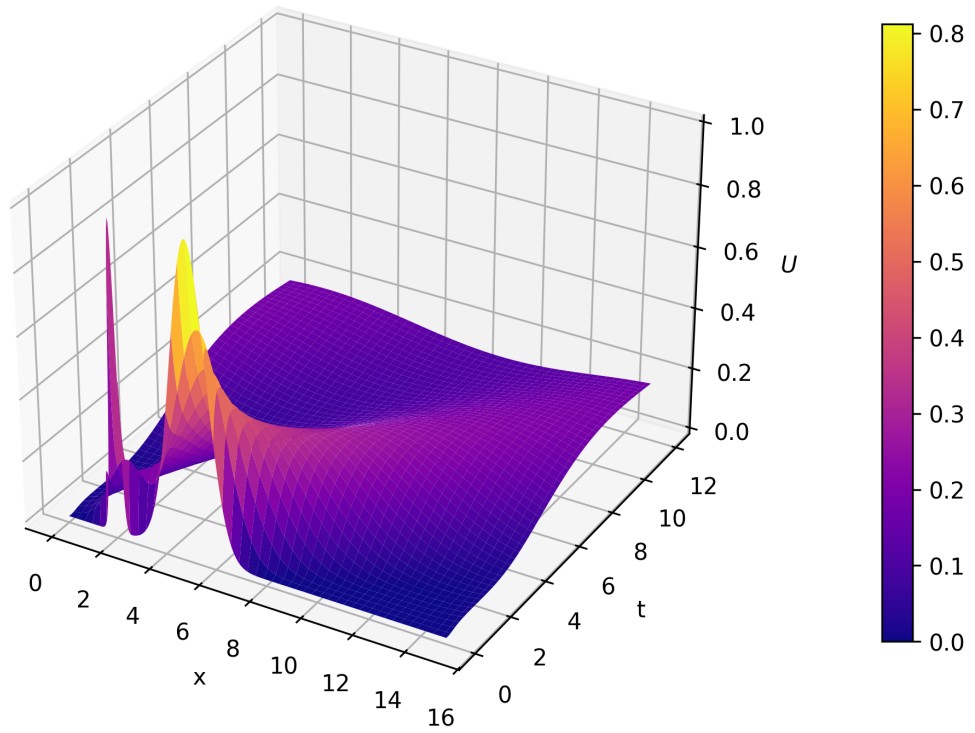


Figura 3: Solução numérica para $Pe = 1$ (Advecção-Difusão)

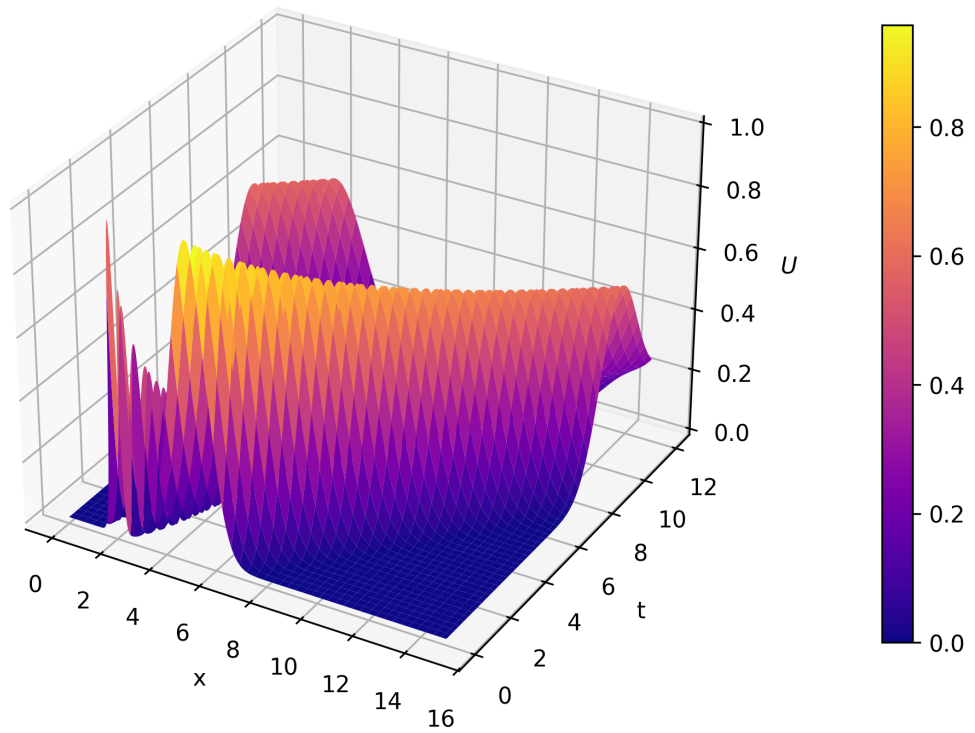


Figura 4: Solução numérica para $Pe = 20 \gg 1$ (Advecção dominante)

2.2 Upwind na derivada espacial

Como o termo que multiplica a primeira derivada espacial é positivo, pelo Upwind utilizamos diferenças regressivas para a derivada espacial, i.e.:

$$\begin{aligned} & \frac{u(x_j, t_{n+1}) - u(x_j, t_n)}{\Delta t} + \mathcal{O}(\Delta t) + \frac{u(x_j, t_n) - u(x_{j-1}, t_n)}{\Delta x} + \mathcal{O}(\Delta x) \\ &= \frac{1}{\mathbb{P}e} \frac{u(x_{j+1}, t_n) - 2u(x_j, t_n) + u(x_{j-1}, t_n)}{\Delta x^2} + \mathcal{O}(\Delta x^2) \end{aligned} \quad (6)$$

que produz o método

$$\frac{U_j^{n+1} - U_j^n}{\Delta t} + \frac{U_j^n - U_{j-1}^n}{\Delta x} = \frac{1}{\mathbb{P}e} \frac{U_{j-1}^n - 2U_j^n + U_{j+1}^n}{\Delta x^2} \quad (7)$$

ou, equivalentemente

$$\begin{aligned} U_j^{n+1} &= \left(\frac{\Delta t}{\mathbb{P}e \Delta x^2} + \frac{\Delta t}{\Delta x} \right) U_{j-1}^n + \left(1 - \frac{2\Delta t}{\mathbb{P}e \Delta x^2} - \frac{\Delta t}{\Delta x} \right) U_j^n + \left(\frac{\Delta t}{\mathbb{P}e \Delta x^2} \right) U_{j+1}^n \\ &= (\alpha + \beta) U_{j-1}^n + (1 - 2\alpha - \beta) U_j^n + \alpha U_{j+1}^n \end{aligned} \quad (8)$$

onde $\alpha = \frac{\Delta t}{\mathbb{P}e \Delta x^2}$ e $\beta = \frac{\Delta t}{\Delta x}$.

Para as condições de contorno periódicas, definimos $U_{N_x+1}^n = U_0^n$, então:
 $j = 0$:

$$U_0^{n+1} = (\alpha + \beta) U_{-1}^n + (1 - 2\alpha - \beta) U_0^n + \alpha U_1^n = (1 - 2\alpha - \beta) U_0^n + \alpha U_1^n + (\alpha + \beta) U_{N_x}^n$$

$j = N_x$:

$$U_{N_x}^{n+1} = (\alpha + \beta) U_{N_x-1}^n + (1 - 2\alpha - \beta) U_{N_x}^n + \alpha U_{N_x+1}^n = \alpha U_0^n + (\alpha + \beta) U_{N_x-1}^n + (1 - 2\alpha - \beta) U_{N_x}^n$$

Portanto, para a forma matricial, teremos

$$\mathbf{U}^{n+1} = \mathbf{A} \mathbf{U}^n \quad (9)$$

$$\text{onde } \mathbf{U}^n = \begin{bmatrix} U_0^n \\ U_1^n \\ U_2^n \\ \vdots \\ U_{N_x}^n \end{bmatrix} \text{ e } \mathbf{A} = \begin{bmatrix} (1 - 2\alpha - \beta) & \alpha & 0 & \cdots & (\alpha + \beta) \\ (\alpha + \beta) & (1 - 2\alpha - \beta) & \alpha & \cdots & 0 \\ 0 & (\alpha + \beta) & (1 - 2\alpha - \beta) & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots \\ \alpha & 0 & \cdots & (\alpha + \beta) & (1 - 2\alpha - \beta) \end{bmatrix}$$

Da condição inicial $u(x, 0) = \exp(-20(x - 2)^2) + \exp(-(x - 5)^2)$, obtemos na forma vetorial

$$\mathbf{U}_j^0 = \begin{bmatrix} \exp\{-20(x_0 - 2)^2\} + \exp\{-(x_0 - 5)^2\} \\ \exp\{-20(x_1 - 2)^2\} + \exp\{-(x_1 - 5)^2\} \\ \vdots \\ \exp\{-20(x_{N_x} - 2)^2\} + \exp\{-(x_{N_x} - 5)^2\} \end{bmatrix}$$

2.2.1 Análise de Estabilidade

Pelo critério de Von Neumann, definimos $U_j^n = e^{i\omega_j}$ e $U_j^{n+1} = \zeta U_j^n = \zeta e^{i\omega_j}$. Logo, do método

$$U_j^{n+1} = (\alpha + \beta)U_{j-1}^n + (1 - 2\alpha - \beta)U_j^n + \alpha U_{j+1}^n$$

temos

$$\begin{aligned}\zeta e^{i\omega_j} &= (\alpha + \beta)e^{i\omega_{j-1}} + (1 - 2\alpha - \beta)e^{i\omega_j} + \alpha e^{i\omega_{j+1}} \\ \zeta &= (\alpha + \beta)e^{-i\omega} + (1 - 2\alpha - \beta) + \alpha e^{i\omega} \\ &= 1 + (e^{-i\omega} + e^{i\omega} - 2)\alpha + (e^{-i\omega} - 1)\beta \\ &= 1 + (2\cos(\omega) - 2)\alpha + (\cos(\omega) - i\sin(\omega) - 1)\beta \\ &= [1 + (2\alpha + \beta)(\cos(\omega) - 1)] - i\beta \sin(\omega)\end{aligned}$$

e o método será estável se $|\zeta|^2 \leq 1$, logo:

$$\begin{aligned}|\zeta|^2 &= [1 + (2\alpha + \beta)(\cos(\omega) - 1)]^2 + [\beta \sin(\omega)]^2 \\ &= 1 + 2(2\alpha + \beta)(\cos(\omega) - 1) + (2\alpha + \beta)^2(\cos(\omega) - 1)^2 + \beta^2 \sin^2(\omega) \\ &= 1 - 4(2\alpha + \beta) \sin^2(\omega/2) + (2\alpha + \beta)^2(-2 \sin^2(\omega/2))^2 + \beta^2 \sin^2(\omega) \\ &= 1 - (8\alpha + 4\beta) \sin^2(\omega/2) + (16\alpha^2 + 16\alpha\beta + 4\beta^2) \sin^4(\omega/2) + \beta^2 \sin^2(\omega) \\ &= 1 - (8\alpha + 4\beta) \sin^2(\omega/2) + (16\alpha^2 + 16\alpha\beta + 4\beta^2) \sin^4(\omega/2) + \beta^2(4 \sin^2(\omega/2) - 4 \sin^4(\omega/2)) \\ &= 1 - (8\alpha + 4\beta - 4\beta^2) \sin^2(\omega/2) + (16\alpha^2 + 16\alpha\beta + 4\beta^2 - 4\beta^2) \sin^4(\omega/2) \\ &\leq 1 - (8\alpha + 4\beta - 4\beta^2) + (16\alpha^2 + 16\alpha\beta)\end{aligned}$$

Portanto $|\zeta|^2 \leq 1 \iff -(8\alpha + 4\beta - 4\beta^2) + (16\alpha^2 + 16\alpha\beta) \leq 0$, logo

$$\begin{aligned}-8\alpha - 4\beta + 4\beta^2 + 16\alpha^2 + 16\alpha\beta &\leq 0 \\ -2\alpha - \beta + \beta^2 + 4\alpha^2 + 4\alpha\beta &\leq 0 \\ -(2\alpha + \beta) + (4\alpha^2 + 4\alpha\beta + \beta^2) &\leq 0 \\ -(2\alpha + \beta) + (2\alpha + \beta)^2 &\leq 0 \\ (2\alpha + \beta)(2\alpha + \beta - 1) &\leq 0\end{aligned}$$

Como $\alpha, \beta > 0$, então $2\alpha + \beta > 0$. Portanto

$$\begin{aligned}|\zeta|^2 \leq 1 &\iff 2\alpha + \beta - 1 \leq 0 \\ 2\frac{\Delta t}{\mathbb{P}e\Delta x^2} + \frac{\Delta t}{\Delta x} &\leq 1 \\ \Delta t \left(\frac{2}{\mathbb{P}e\Delta x^2} + \frac{1}{\Delta x} \right) &\leq 1 \\ \Delta t \left(\frac{2 + \mathbb{P}e\Delta x}{\mathbb{P}e\Delta x^2} \right) &\leq 1 \\ \Delta t \left(\frac{\frac{2}{\mathbb{P}e} + \Delta x}{\Delta x^2} \right) &\leq 1 \\ \therefore \Delta t &\leq \frac{\Delta x^2}{\frac{2}{\mathbb{P}e} + \Delta x}\end{aligned}$$